Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

We are given :

- Cost to plant the tree : $ 9 0 (one -time cost)
- Each year , the tree produces 7 le mons
- Each lemon sells for $ 1 . 5 0
- Annual cost to water and feed the tree : $ 3

We need to find ** how many years it will take before Carlos starts earning money ** âĢĶ that is , when the ** annual income from le mons exceeds the annual expenses ** (water ing and feeding), and the ** net profit becomes positive ** .

But note : the initial $ 9 0 cost is a one -time cost . So Carlos starts earning money * after * the first year , when the ** annual net profit ** (re venue from le mons minus costs) covers the initial investment .

Let âĢĻs compute the ** net annual profit ** from the lemon tree :

Annual revenue from le mons :
$$
7 \text { le mons } \times \$ 1 . 5 0 = \$ 1 0 . 5 0
$$

Annual cost (water ing and feeding):
$$
\$ 3
$$

Net annual profit :
$$
\$ 1 0 . 5 0 - \$ 3 = \$ 7 . 5 0 \text { per year }
$$

So , each year after planting , Carlos earns $ 7 . 5 0 net profit .

Now , the total profit after $ n $ years is :

$$
\text { Total profit } = 7 . 5 n - 9 0
$$

We want to find the smallest integer $ n $ such that the ** total profit becomes positive **, i .e ., when he starts earning money (not just breaking even):

$$
7 . 5 n - 9 0 > 0
$$

S olve for $ n $:

$$
7 . 5 n > 9 0 \\
n > \frac { 9 0 }{ 7 . 5 } \\
n > 1 2
$$

So , after ** 1 2 years **, he has exactly broken even .

After ** 1 3 years **, he starts earning money (since the net profit becomes positive).

But the question is : ** How many years will it take before he starts earning money ? **

This means : when does he begin to make a profit ? After 1 3 years .

But let âĢĻs double -check :

- After 1 2 years : total profit = $ 7 . 5 \times 1 2 - 9 0 = 9 0 - 9 0 = 0 $ âĨĴ break -even
- After 1 3 years : $ 7 . 5 \times 1 3 = 9 7 . 5 $, so profit = $ 9 7 . 5 - 9 0 = 7 . 5 $ âĨĴ positive profit

So , he starts earning money ** after 1 3 years ** .

Therefore , the answer is :

< answer > 1 3 </ answer >